

Week 5 — Classic IP Models and Networks

OL7014 Integer Programming • Instructor: Prof. H. Park • Week of Sep 21, 2026

1. The LP relaxation — and the question it leaves

LP relaxation: delete the integrality line — optimize over P instead of $X = P \cap \mathbb{Z}^n$

$$(\text{IP}) \max_x \{c^\top x : Ax \leq b, x \in \mathbb{Z}_+^n\} \longrightarrow (\text{LP}) \max_x \{c^\top x : Ax \leq b, x \geq 0\}$$

larger feasible set $\Rightarrow$ always optimistic: $z_{\text{LP}} \geq z_{\text{IP}}$ (max), $z_{\text{LP}} \leq z_{\text{IP}}$ (min)

today: when $z_{\text{LP}} = z_{\text{IP}}$?

KEY one line, $\in \mathbb{Z}$, is what makes an IP hard — so a model where deleting it costs nothing is worth recognizing on sight.

2. Assignment Problem

demo ip-games / assignment — <https://hankpark0706.github.io/ip-games/assignment-matching.html>

4 cadets (A–D), 4 branches (Army, Navy, Air Force, Marines), one each — also officers to billets, units to sectors: the **assignment problem**

c_{ij} = fit score of cadet i in branch j (the game's 1–5 table); the decision is a *table*: $X \in \{0, 1\}^{4 \times 4}$, $x_{ij} = 1$ iff cadet i takes branch j

rows sum to 1 (each cadet once), columns sum to 1 (each branch once) — week 2's BP:

$$\begin{aligned} \max_X \quad & \langle C, X \rangle \\ \text{s.t.} \quad & X\mathbf{1} = \mathbf{1}, \quad X^\top \mathbf{1} = \mathbf{1} \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

read the symbols: $\langle C, X \rangle = \sum_{i,j} c_{ij}x_{ij}$, $\mathbf{1}$ = all-ones vector

canonical form (week 2), 3 cadets (A, B, C), 3 branches (Army, Navy, Air Force) — flatten X row by row:

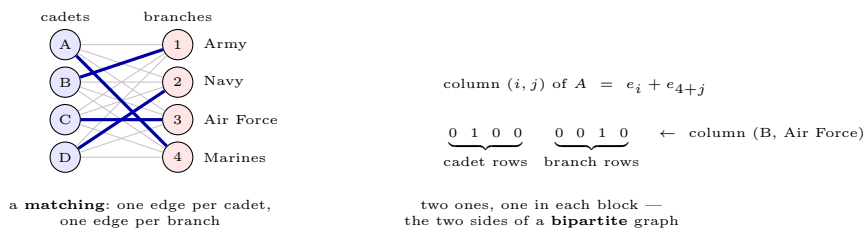
$$x = \text{vec}(X) = (x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33}), \quad c = \text{vec}(C)$$

$$\max_x c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \in \{0, 1\}^9$$

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

cadet A once
cadet B once
cadet C once
Army filled once
Navy filled once
Air Force filled once

A is $2n \times n^2$ — one row per constraint, one column per (cadet, branch) pair: here 6×9 , the game 8×16



KEY the LP relaxation gives the same solution as the IP — why?

takeaway delete assignment's integrality line and the answer does not move. the reason has a name — *total unimodularity* — coming up.

3. Transportation Problem

demo `ip-games / transportation` — two facilities, three divisions — the 2×3 case below. <https://hankpark0706.github.io/ip-games/transportation.html>

supply s_i at source i , demand d_j at sink j (assignment: all = 1) $\Rightarrow$ matching becomes **flow**

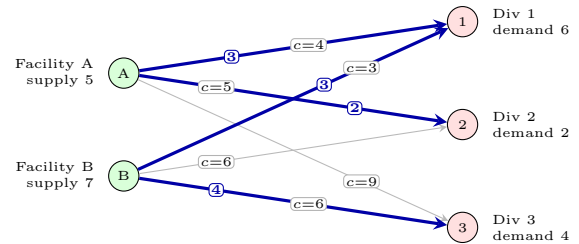
x_{ij} = amount shipped $i \rightarrow j$

$$\begin{aligned} \min_x \quad & \sum_{i=1}^n \sum_{j=1}^k c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^k x_{ij} \leq s_i \quad (i \in [n]) \quad \text{ship no more than you hold} \\ & \sum_{i=1}^n x_{ij} \geq d_j \quad (j \in [k]) \quad \text{every sink gets what it needs} \\ & x_{ij} \in \mathbb{Z}_+ \end{aligned}$$

$$A = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$b = \begin{pmatrix} s_1 \\ s_2 \\ d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

facility A ships s_1
facility B ships s_2
division 1 gets d_1
division 2 gets d_2
division 3 gets d_3



the game's board: $b = (5, 7, 6, 2, 4)$
blue: the cheapest plan, amounts shipped — $3 \cdot 4 + 2 \cdot 5 + 3 \cdot 3 + 4 \cdot 6 = 55$

assignment's pattern, now $(n+k) \times nk = 5 \times 6$ — only b changed: s, d in place of $\mathbf{1}$

relax: $x_{ij} \in \mathbb{Z}_+ \rightarrow x_{ij} \geq 0$

assignment is the special case of transportation problem: $k = n$, $s_i = d_j = 1$ — row sums cap x_{ij} at 1, and section 4 puts the optimum on a *corner* — a permutation matrix

takeaway transportation = assignment with quantities: supplies left, demands right, flows between. an IP you may hand to an LP solver whenever the data is whole.

4. Total unimodularity — why the LP told the truth

gap on week 1's toy: $22.5 - 19$. gap on assignment, transportation: 0. same solver $\Rightarrow$ the difference is in the *matrix*

Def. A is **totally unimodular (TU)**: every square submatrix has determinant 0, +1, or -1

(any k rows $\times$ any k columns, every k) — a property of A alone: no b , no c

determinant: $\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ — square matrix in, one number out

$$1 \times 1 : \det(a) = a \quad \quad 2 \times 2 : \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$3 \times 3 : \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

$$\det(8) = 8, \det(-1) = -1 \quad \det \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} = 1 \cdot 1 - 3 \cdot 3 = -8 \text{ (week 1's toy)}, \quad \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1 \cdot 1 - 1 \cdot 0 = 1$$

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = 1(1 \cdot 1 - 1 \cdot 0) - 1(0 \cdot 1 - 1 \cdot 1) + 0 = 1 + 1 = 2 \quad \text{— entries all 0, 1, yet } \det = 2: \text{ not TU}$$

$k = 1$ is **free**: a 1×1 submatrix is one entry $\Rightarrow$ **every entry of a TU matrix is 0, ± 1**

Prop. (Hoffman–Kruskal) A TU and b integer $\Rightarrow$ every corner of $P = \{x : Ax \leq b, x \geq 0\}$ is integer (same for $Ax = b$)

$\Rightarrow$ the LP, if it has an optimum, has an *integer* one: **the relaxation solves the IP**

Prop. (TU-preserving; Wolsey, Ch. 3) A is TU $\Leftrightarrow \begin{pmatrix} A \\ I \end{pmatrix}$ is TU

why: a row of I holds a single 1 — expand the determinant along it, and what is left is 0 or $\pm$ a smaller one

so **bounds are free**: $x_j \leq u_j$ is a row of I — the same kind of row as the $x \geq 0$ already inside P

KEY TU matrix A and integer b guarantee $z_{LP} = z_{IP}$.

5. Recognizing it — two dimensions, then the checklist

week 1's objective $\max 4x + 5y$, $x, y \geq 0$, change *nothing but* A :

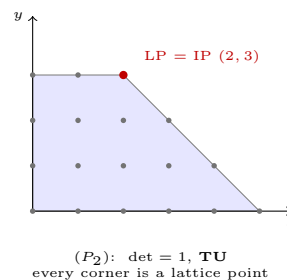
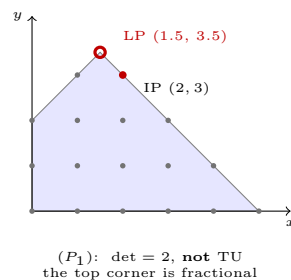
$$(P_0) \begin{cases} x + 3y \leq 10 \\ 3x + y \leq 10 \end{cases} \quad (P_1) \begin{cases} x + y \leq 5 \\ -x + y \leq 2 \end{cases} \quad (P_2) \begin{cases} x + y \leq 5 \\ y \leq 3 \end{cases}$$

(P_0) = week 1's toy. (P_1) : two teams fly ≤ 5 sorties between them, B at most two *more than* A. (P_2) : flat cap, B flies ≤ 3

	A	largest det	TU?	LP optimum	IP optimum
(P_0)	$\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$	8	no — there is a 3	(2.5, 2.5), 22.5	(1, 3), 19
(P_1)	$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$	2	no — det = 2	(1.5, 3.5), 23.5	(2, 3), 23
(P_2)	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	1	yes	(2, 3), 23	(2, 3), 23

(adding the rows $-x \leq 0$, $-y \leq 0$ changes nothing here)

(P_1) vs (P_2) : **one entry** apart. $\det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 2 \Rightarrow$ a division by 2 $\Rightarrow$ the 1.5. $-1 \rightarrow 0$: $\det = 1$, corners snap to integers



two traps. 1. entries all $0, \pm 1 \not\Rightarrow$ TU — see (P_1) . 2. TU belongs to the *whole* matrix: neither row of (P_1) is guilty alone

checklist (sufficient, not necessary):

- (i) every entry is $0, \pm 1$
- (ii) every column has at most two nonzeros
- (iii) rows split into two sets (M_1, M_2) , $M_2 = \emptyset$ allowed, so that **every column j with two nonzeros balances**:
$$\sum_{i \in M_1} a_{ij} = \sum_{i \in M_2} a_{ij}$$

$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} +1 & +1 & 0 \\ -1 & 0 & +1 \\ 0 & -1 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$
TU — assignment (§2)	TU — transportation (§3)	TU — MCNF (§7)	TU	not TU (§4)
M_1 = cadet rows M_2 = branch rows	M_1 = facility rows M_2 = division rows	M_1 = all rows, $M_2 = \emptyset$ $(+1) + (-1) = 0$	M_1 = row 1, M_2 = rows 2, 3 col. 3 exempt: one nonzero	no split: an odd cycle

$\Rightarrow$ their matrices are TU, so for integer b Hoffman–Kruskal (§4) applies: **assignment and transportation solve as LPs**

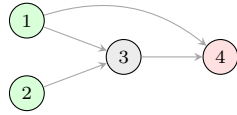
6. Graphs — the notation every model from here on lives on

sections 2 and 3 used it without saying so; from here to week 12 every model is written on a graph

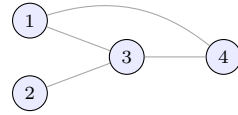
$G = (V, E)$: V = a finite set of **nodes** (the *things* — assignment: cadets and branches; transportation: facilities and troops), $E \subseteq V \times V$ = a set of **arcs** (the *connections* — a permitted pairing; a road)

$|V| = m$ nodes = rows of A ; $|E| = n$ arcs = columns of A — **one decision x_{ij} per arc**

	an arc is	drawn as
directed	an ordered pair (i, j) : leaves i (tail), enters j (head); $(i, j) \neq (j, i)$	arrow $i \rightarrow j$
undirected	a set $\{i, j\}$, no direction — an <i>edge</i>	line $i - j$



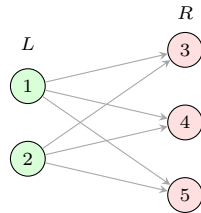
directed: arrows — $(1, 3)$ leaves 1, enters 3
node 3 is a hub, flow in and out — section 7



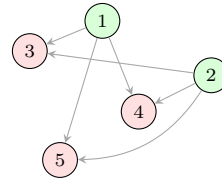
undirected: the same four nodes, lines — $\{1, 3\} = \{3, 1\}$
a road with no one-way sign; no tail, no head, no hub

bipartite — a property of the *node set*, not of an arc (a bipartite graph can be directed or not): $V = L \cup R$, $L \cap R = \emptyset$, every arc joins L to R — none inside L , none inside R . usually drawn as two columns, but the columns are not the definition

the graph above is **not** bipartite: $1 \rightarrow 3 \rightarrow 4$ and $1 \rightarrow 4$ make a triangle, and no two-way split of $\{1, 3, 4\}$ keeps every arc across



bipartite: every arc $L \rightarrow R$
sections 2, 3 — no hubs



same six arcs, nodes shuffled: **still bipartite**
 L, R are the *colors*, not the columns

the classics, read as graphs:

model	L	R	E	x_{ij} on arc (i, j)
assignment (§2)	n cadets	n branches	every pair: n^2 arcs (the game: 16)	$\{0, 1\}$: paired or not
transportation (§3)	n facilities	k divisions	every pair: nk arcs (§3's arrows)	$\mathbb{Z}_+$: crates shipped

KEY a graph is two lists, nodes and arcs. a model on it is one row per node, one decision per arc — sections 2 and 3 were bipartite graphs before the word appeared.

7. Minimum cost network flow (MCNF) — the family worth recognizing

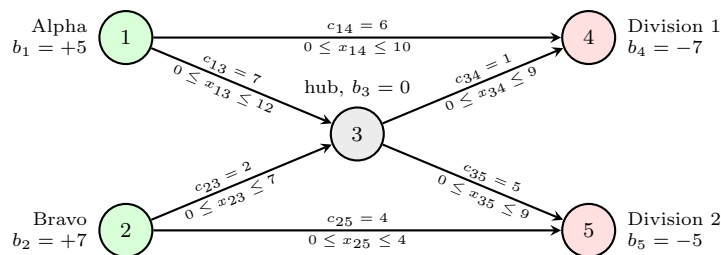
demo [ip-games / min-cost-flow](https://hankpark0706.github.io/ip-games/min-cost-flow.html) — push flow through a network by hand, then let the solver do it. <https://hankpark0706.github.io/ip-games/min-cost-flow.html>

MCNF drops the bipartite restriction: any directed graph $G = (V, E)$, $|V| = m$ nodes, arcs $(i, j) \in E$

node i : b_i = net supply = outflow — inflow **arc (i, j) :** cost c_{ij} , bounds $l_{ij} \leq x_{ij} \leq u_{ij}$ **decision:** x_{ij} = flow (whole units $\Rightarrow$ integer)

$b_i > 0$	node i supplies	source
$b_i < 0$	node i demands	destination
$b_i = 0$	flow only passes through	transshipment, a hub

$$\begin{aligned}
 \min_x \quad & \sum_{(i,j) \in E} c_{ij} x_{ij} \\
 \text{s.t.} \quad & \underbrace{\sum_{j: (i,j) \in E} x_{ij} - \sum_{j: (j,i) \in E} x_{ji}}_{\text{one row per node — flow conservation}} = b_i \quad (\forall i \in V) \\
 & l_{ij} \leq x_{ij} \leq u_{ij}, \quad x_{ij} \in \mathbb{Z}_+
 \end{aligned}$$



the game's map — each road carries its cost per truckload and the bounds on its one decision

$$\begin{pmatrix} A \\ I \end{pmatrix} = \begin{matrix} & \begin{matrix} \text{arc} & \text{arc} & \text{arc} & \text{arc} & \text{arc} & \text{arc} \\ (1,3) & (1,4) & (2,3) & (2,5) & (3,4) & (3,5) \end{matrix} \\ \begin{matrix} \text{node 1} \\ \text{node 2} \\ \text{node 3} \\ \text{node 4} \\ \text{node 5} \\ \text{capacity (1,3)} \\ \text{capacity (1,4)} \\ \text{capacity (2,3)} \\ \text{capacity (2,5)} \\ \text{capacity (3,4)} \\ \text{capacity (3,5)} \end{matrix} & \begin{pmatrix} +1 & +1 & 0 & 0 & 0 & 0 \\ 0 & 0 & +1 & +1 & 0 & 0 \\ -1 & 0 & -1 & 0 & +1 & +1 \\ 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix} \left. \begin{matrix} = 5 \\ = 7 \\ = 0 \\ = -7 \\ = -5 \\ \leq 12 \\ \leq 10 \\ \leq 7 \\ \leq 4 \\ \leq 9 \\ \leq 9 \end{matrix} \right\} \begin{matrix} A \text{ incidence matrix, } = b \\ \\ \\ \\ \\ I \text{ capacities, } \leq u \end{matrix}$$

A is **TU** — every column holds one $+1$ (tail) and one -1 (head): section 5's MCNF example, $M_1 = \text{all rows}$, $M_2 = \emptyset$

$\begin{pmatrix} A \\ I \end{pmatrix}$ is **TU** — section 4's Prop (TU-preserving): the capacity rows are I , so *bounds are free*

(undirected: two $+1$ s need a bipartite split — the **matching LP** is integral only on bipartite graphs)

$\text{rank } A = |V| - 1 = 4$: one conservation row is always redundant (rows sum to zero)

Prop. integer $\{b_i\}$ and integer bounds $\{l_{ij}\}, \{u_{ij}\} \Rightarrow$ every extreme point is integral: delete $x \in \mathbb{Z}_+$, solve the LP, **no branching, ever**

one model, four classics:

problem	is MCNF with
transportation	sources $b_i = s_i > 0$, destinations $b_j = -d_j$
assignment	transportation with every $b_i = +1$, $b_j = -1$ (equally many of each)
shortest path	$b_s = +1$, $b_t = -1$, $b_i = 0$ otherwise
max flow	all $b_i = 0$; add an artificial arc (t, s) and maximize x_{ts}

sections 2 and 3 were the same model twice — both bipartite networks, the case with no hubs. the game is the first with a hub

KEY recognize an **IP** as a network and it is solved: **one row per node, one column per arc, ± 1 entries** — **TU by construction, so the relaxation is free.**

next lecture facility location: transportation with a *switch* $x_i \in \{0, 1\}$ on each facility, price f_i — which facilities should exist? then the LAP: MCNF where the supplies b_i are *chosen*, not given — which facilities open, how much each holds — and the LP relaxation starts to lie. **Homework 2 is out** (due 10/13).